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上海卷 春考

阅读 C

化石：判断岩石年代的依据

说明文 · 地质与古生物 · 全文 416 词 · 逐句精读

¹ To **tell the age of** most **layered** rocks, **scientists** study the **fossils** these rocks **contain**.

要判断大多数层状岩石的年代，科学家会研究这些岩石中所含的化石。

tell the age of 判断……的年代

layered adj. 分层的；层状的

scientist n. 科学家

fossil n. 化石

contain v. 包含；含有

² Fossils **provide** important **evidence** to help **determine** what happened in Earth **history** and when it happened.

化石提供了重要证据，有助于确定地球历史上发生过什么以及发生在何时。

provide v. 提供

evidence n. 证据

determine v. 确定；查明

history n. 历史

³ The word “fossil” **makes** many people **think of** **dinosaurs**.

“化石”这个词让许多人想到恐龙。

make...think of 使……想到

dinosaur n. 恐龙

⁴ Dinosaurs are now **featured** in books, movies, and television **programs**, and the **bones** of some large dinosaurs are **on display** in many **museums**.

如今，恐龙出现在书籍、电影和电视节目中，一些大型恐龙的骨骼还在许多博物馆里展出。

feature v. 以……为主要内容；使出现（在书、电影等中）

program n. 节目

bone n. 骨头；骨骼

on display 展出；陈列

museum n. 博物馆

⁵ These **reptiles** were **dominant** animals on Earth for **well over** 100 million years from the Late **Triassic** through the Late **Cretaceous**.

从三叠纪晚期到白垩纪晚期，在远超1亿年的时间里，这些爬行动物一直是地球上的优势动物。

reptile n. 爬行动物

dominant adj. 占优势的；占主导地位的

well over 远远超过

Triassic n. 三叠纪

Cretaceous n. 白垩纪

⁶ Many dinosaurs were **quite** small, but by the middle of the **Mesozoic Era**, some **species** **weighed** **as much as** 80 **tons**.

许多恐龙体型相当小，但到了中生代中期，有些物种的体重达80吨之多。

quite adv. 相当；颇

Mesozoic adj. 中生代的

era n. 时代；（地质）代

species n. 物种

weigh v. 重（若干）

as much as 多达

ton n. 吨

⁷ By around 65 **million** years ago all dinosaurs were **extinct**.

到大约6500万年前，所有恐龙都已灭绝。

million num. 百万

extinct adj. 灭绝的

⁸ **In spite of** all of the **interest** in dinosaurs, they **form** only a small **fraction** of the millions of species that live and have lived on Earth.

尽管人们对恐龙兴趣浓厚，但在地球上现存和曾经存在过的数百万个物种中，恐龙只占很小一部分。

in spite of 尽管

interest n. 兴趣；关注

form v. 构成；组成

fraction n. 小部分；一点儿

- 9 **The great bulk of** the fossil **record** is **dominated** by fossils of animals with **shells** and **microscopic** **remains** of plants and animals, and these remains are **widespread** in **sedimentary** rocks.

化石记录的绝大部分是有壳动物的化石和动植物的微体遗存，这些遗存广泛分布于沉积岩中。

the great bulk of的绝大部分

record n. 记录

dominate v. 在……中占主导地位；支配

shell n. 壳；贝壳

microscopic adj. 微小的；需用显微镜才能看见的

remains n. 遗体；遗存

widespread adj. 分布广的；普遍的

sedimentary adj. 沉积的

10 It is these fossils that are studied by most paleontologists.

大多数古生物学家研究的正是这些化石。

it is...that 正是……（强调句）

paleontologist n. 古生物学家

- 11 In the late **eighteenth** and early nineteenth **centuries**, the English **geologist** and **engineer** William Smith and the French paleontologists Georges Cuvier and Alexandre Brongniart **discovered** that rocks of the same age may contain the same fossils even when the rocks are **separated** by long **distances**.

18世纪末和19世纪初，英国地质学家兼工程师威廉·史密斯以及法国古生物学家乔治·居维叶和亚历山大·布隆尼亚尔发现，即使相隔遥远，同一年代的岩石也可能含有相同的化石。

eighteenth num. 第十八

century n. 世纪

geologist n. 地质学家

engineer n. 工程师

discover v. 发现

separate v. 分隔；分开

distance n. 距离

¹² They **published** the first **geologic** maps of large **areas** on which rocks containing **similar** fossils were shown.

他们出版了首批覆盖大片区域的地质图，图上标出了含有相似化石的岩石。

publish v. 出版；发表

geologic adj. 地质的

area n. 地区；区域

similar adj. 相似的

¹³ By careful **observation** of the rocks and their fossils, these men and other geologists **were able to** **recognize** rocks of the same age on **opposite** sides of the **English Channel**.

通过仔细观察岩石及其所含化石，这些人和其他地质学家得以辨认出英吉利海峡两岸同一年代的岩石。

observation n. 观察

be able to 能够

recognize v. 辨认出；识别

opposite adj. 对面的；相反的

English Channel 英吉利海峡

¹⁴ William Smith was able to apply his knowledge of fossils in a very practical way.

威廉·史密斯得以把自己的化石知识运用到非常实际的用途上。

apply v. 运用；应用

knowledge n. 知识

practical adj. 实际的；实用的

¹⁵ He was an engineer building **canals** in England, which has lots of **vegetation** and few **surface exposures** of rock.

他是在英格兰修建运河的工程师，而英格兰植被茂盛，地表很少有裸露的岩石。

canal n. 运河

vegetation n. 植被；（统称）植物

surface n. 地表；表面

exposure n. 露出；（岩石的）露头

¹⁶ He needed to know what rocks he could **expect** to find on the **hills** through which he **had to** build a canal.

他需要知道，在运河必须穿过的那些山丘上，自己预计会找到哪些岩石。

expect v. 预料；预计

hill n. 小山；丘陵

have to 不得不；必须

17 Often he could tell what kind of rock **was likely to** be below the surface by **examining** the fossils that had **eroded** from the rocks of the **hillside** or by **digging** a small hole to find fossils.

他查看从山坡岩石上侵蚀脱落的化石，或者挖个小洞找化石，往往就能判断出地表以下可能是哪种岩石。

be likely to 可能

examine v. 检查；查看

erode v. 侵蚀；（因侵蚀而）脱落

hillside n. 山坡

dig v. 挖；掘

¹⁸ Knowing what rocks to expect **allowed** Smith **to** **estimate** **costs** and determine what **tools** were needed for the job.

知道会遇到哪些岩石，史密斯就能估算成本，并确定这项工程需要哪些工具。

allow...to 使……能够；允许……做

estimate v. 估算；估计

cost n. 成本；费用

tool n. 工具

¹⁹ Smith and others knew that the **succession** of **life forms** **preserved** as fossils is **useful** for understanding how and when the rocks **formed**.

史密斯等人知道，保存为化石的各种生命形式，其演替顺序有助于了解这些岩石如何形成、何时形成。

succession n. 演替；连续的一系列

life form 生命形式；生物

preserve v. 保存

useful adj. 有用的；有益的

form v. 形成

²⁰ Only later did scientists develop a theory to explain that succession.

直到后来，科学家们才提出了解释这种演替顺序的理论。

only later did 直到后来才（倒装）

develop v. 提出；发展出

theory n. 理论

explain v. 解释